

2.3 Trigonometric Substitutions

Motivating Example:

Suppose we wish to find $\int \sqrt{a^2 - x^2} \, dx$ where a is some positive real number.

One question we can ask about this integral is: Is there a substitution we can make that will simplify the integration process?

A good guess would be to make the substitution $x = a \sin(\theta)$ because if we make this substitution we have:

$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2(\theta)} = \sqrt{a^2(1 - \sin^2(\theta))} = \sqrt{a^2 \cos^2(\theta)} = a|\cos(\theta)|$ and $dx = a \cos(\theta) d\theta$ (these equations seem like they might yield an easier integral).

Subtle issues may arise with making the substitution $x = a \sin(\theta)$. In particular one could ask: For each possible value of x is there a unique θ which yields that value of x ?

In general, if the answer to this question is no, then there are several issues we could run into:

- It may be difficult to write our final answer in terms of x .
- We may end up getting expressions for θ that are annoying to integrate (In this example we got an absolute value of cosine involved).
- If we are dealing with a definite integral we may have several choices for the bounds when the integral is in terms of θ .

To avoid these potential issues and make our calculations simpler, we restrict θ so that $x(\theta) = a \sin(\theta)$ is a one to one function (*Note: we are thinking of x as a function of θ*).

In this example we would restrict θ to be in the interval: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. If we do this we have:

$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2(\theta)} = \sqrt{a^2(1 - \sin^2(\theta))} = \sqrt{a^2 \cos^2(\theta)} = a \cos(\theta)$ and $x = a \sin(\theta) \Leftrightarrow \theta = \sin^{-1}\left(\frac{x}{a}\right)$. We

note that the restrictions on θ allowed us to drop the absolute value sign and solve for θ in terms of x . We will return to an example similar to this later.

Now, we will use the above ideas to determine what type of trigonometric substitution we will make in different situations.

Table of Trigonometric Substitutions:

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2(\theta) = \cos^2(\theta)$
$\sqrt{a^2 + x^2}$	$x = a \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2(\theta) = \sec^2(\theta)$
$\sqrt{x^2 - a^2}$	$x = a \sec(\theta)$ where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$	$\sec^2(\theta) - 1 = \tan^2(\theta)$

Example 1 (you try): Suppose $x = a \sec(\theta)$ where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$ and a is positive. Write $\sqrt{x^2 - a^2}$ in terms of θ (fully simplify your answer).

We note that if $x = a \sec(\theta)$, then $\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2(\theta) - a^2} = a |\tan(\theta)|$. We note that $\tan(\theta)$ is positive when $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$. So, this simplifies further to $a \tan(\theta)$.

Note: This example provides some rationale as to why θ is restricted the way that it is in the third row of the table. Also note that when we have $x = a \sec(\theta)$ we cannot solve for x in terms of the inverse secant because the range of inverse secant does not match the restrictions.

Example 2: Find $\int \frac{\sqrt{9-x^2}}{x^2} dx$.

Following the table above we let $x = 3 \sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. This implies $\theta = \sin^{-1}\left(\frac{x}{3}\right)$ and $dx = 3 \cos(\theta) d\theta$.

Now, with this substitution we have: $\frac{\sqrt{9-x^2}}{x^2} = \frac{\sqrt{9-9\sin^2(\theta)}}{9\sin^2(\theta)} = \frac{\sqrt{9(1-\sin^2(\theta))}}{9\sin^2(\theta)} = \frac{3\cos(\theta)}{9\sin^2(\theta)} = \frac{\cos(\theta)}{3\sin^2(\theta)}$.

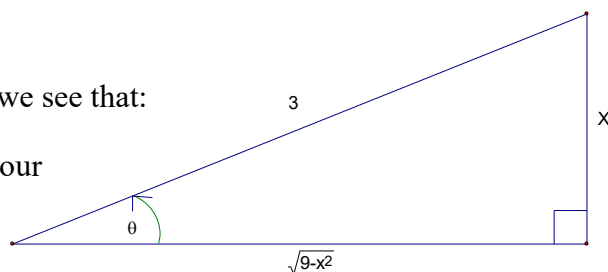
This means that the integral becomes:

$$\begin{aligned} \int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{\cos(\theta)}{3\sin^2(\theta)} 3\cos(\theta) d\theta \\ &= \int \cot^2(\theta) d\theta = \int (\csc^2(\theta) - 1) d\theta \\ &= -\cot(\theta) - \theta + C. \end{aligned}$$

Now, we must write our final answer in terms of x . To do this we can use trig identities or a reference triangle. In class I will lean towards using a reference triangle. In this case, since $\sin(\theta) = \frac{x}{3}$ the reference triangle is (*Note: the reference triangle would be the same in the first or fourth quadrant*):

From the reference triangle and what we have shown we see that:

$-\cot(\theta) - \theta + C = \frac{-\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$ which is our answer.



Example 3: Find $\int \frac{dx}{\sqrt{x^2 - 4}}$.

Following the table we let $x = 2\sec(\theta)$ where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$. This implies $dx = 2\sec(\theta)\tan(\theta)d\theta$

Now, with this substitution we have: $\frac{1}{\sqrt{x^2 - 4}} = \frac{1}{\sqrt{4\sec^2(\theta) - 4}} = \frac{1}{2\tan(\theta)}$.

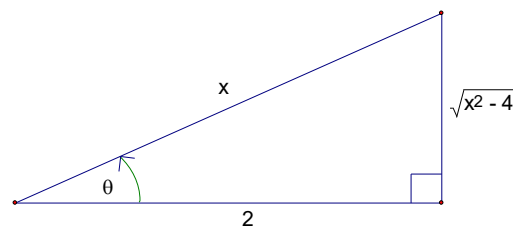
This means that the integral becomes:

$$\int \frac{dx}{\sqrt{x^2 - 4}} = \int \frac{1}{2\tan(\theta)} 2\sec(\theta)\tan(\theta) d\theta = \int \sec(\theta) d\theta = \ln|\sec(\theta) + \tan(\theta)| + C.$$

Now, we must write our final answer in terms of x . In this case, since $\sec(\theta) = \frac{x}{2}$ the reference triangle (in the first quadrant) is: (Note: the reference triangle yields the same secant and tangent regardless if it is in the first or third quadrant):

From the reference triangle and what we have shown we see that:

$$\ln|\sec(\theta) + \tan(\theta)| + C = \ln\left|\frac{x}{2} + \frac{\sqrt{x^2 - 4}}{2}\right| + C \text{ which is our answer.}$$



Note: It is also possible to do this problem with hyperbolic trig functions. See page 481 in your textbook if you are interested in seeing this.

Example 4: Find $4 \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\frac{1}{4} - x^2} dx$

To evaluate this integral we let $x = \frac{1}{2}\sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. This implies $\theta = \sin^{-1}\left(\frac{x}{(1/2)}\right)$ and

$dx = \frac{1}{2}\cos(\theta)d\theta$. The limits of integration for theta are found by plugging $-\frac{1}{2}$ & $\frac{1}{2}$ in for x

$$\therefore \sin^{-1}\left(\frac{-1/2}{1/2}\right) = \frac{-\pi}{2} \text{ and } \sin^{-1}\left(\frac{1/2}{1/2}\right) = \frac{\pi}{2}.$$

$$4 \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\frac{1}{4} - x^2} dx = 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1}{4} - \left(\frac{1}{2}\sin(\theta)\right)^2} dx = 4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1}{4}(1 - \sin^2(\theta))} \cdot \frac{1}{2}\cos(\theta) d\theta$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}\sqrt{\cos^2(\theta)} \cdot \cos(\theta) d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(\theta) d\theta = \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [1 + \cos(2\theta)] d\theta = \frac{1}{2} \left[\left(\theta + \frac{1}{2}\sin(2\theta) \right) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\theta + \frac{1}{2}\sin(2\theta) \right) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{1}{2}\sin(\pi) \right) - \left(-\frac{\pi}{2} + \frac{1}{2}\sin(-\pi) \right) \right] = \frac{1}{2}[\pi] = \frac{\pi}{2}$$

Example 5 (you try):

(a) Find the exact value of: $\int_0^1 \frac{dx}{(x^2+1)^2}$. *Hint:* $\int_0^1 \frac{dx}{(x^2+1)^2} = \int_0^1 \frac{dx}{(\sqrt{x^2+1})^4}$

To evaluate this integral we let $x = \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. This implies $\theta = \tan^{-1}(x)$ and $dx = \sec^2(\theta)d\theta$.

Thus, the new limits of integration will be: $\theta(0) = \tan^{-1}(0) = 0$ and $\theta(1) = \tan^{-1}(1) = \frac{\pi}{4}$.

Now, with this substitution we have: $\frac{1}{(\sqrt{x^2+1})^4} = \frac{1}{(\sqrt{\tan^2(\theta)+1})^4} = \frac{1}{\sec^4(\theta)} = \cos^4(\theta)$.

Now, we calculate our answer:

$$\begin{aligned} \int_0^1 \frac{dx}{(\sqrt{x^2+1})^4} &= \int_0^{\frac{\pi}{4}} \cos^4(\theta) \sec^2(\theta) d\theta = \int_0^{\frac{\pi}{4}} \cos^2(\theta) d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{2}(1 + \cos(2\theta)) d\theta = \left(\frac{1}{2}\theta + \frac{1}{4}\sin(2\theta) \right) \Big|_0^{\frac{\pi}{4}} = \frac{\pi}{8} + \frac{1}{4}. \end{aligned}$$

(b) In no more than three sentence explain how you would find $\int \frac{x}{\sqrt{1+x^2}} dx$.

A simple Calculus 1 u -substitution will work. If we let $u = 1+x^2$.

Note: Part (b) of this example emphasizes a very important point. Specifically, we do not want to use trigonometric substitution if a simpler method of integration will be successful.

The following example shows that completing the square can be a valuable tool when using trigonometric substitution.

Example 6

Consider $\int_{-3}^1 \frac{x}{\sqrt{3-2x-x^2}} dx$. Complete the square and perform a substitution to set up (do not evaluate) an equivalent integral to which trigonometric substitution can be applied.

We calculate that $3-2x-x^2 = -(x^2+2x)+3 = -(x^2+2x+1)+3+1 = 4-(x+1)^2$.

So, we have that $\int_{-3}^1 \frac{x}{\sqrt{3-2x-x^2}} dx = \int_{-3}^1 \frac{x}{\sqrt{4-(x+1)^2}} dx$.

Now, we let $u = x+1$ which implies $x = u-1$ and $dx = du$.

Also, to find the bounds on u we plug in -3 and 1 in for x and obtain $-3+1 = -2$ for the lower bound and $1+1 = 2$ for the upper bound.

So, the integral becomes: $\int_{-3}^1 \frac{x}{\sqrt{4-(x+1)^2}} dx = \int_{-2}^2 \frac{u-1}{\sqrt{4-u^2}} du$, and we now have an integral to which we can apply trig substitution.